NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

MIL-PRF-27/71C
23 July 1990
SUPERSEDING
MIL-T-27/71B
11 July 1979

PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, AUDIO FREQUENCY TF5S21ZZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformers described herein shall consist of this document and the latest issue of Specification MIL-T-27.

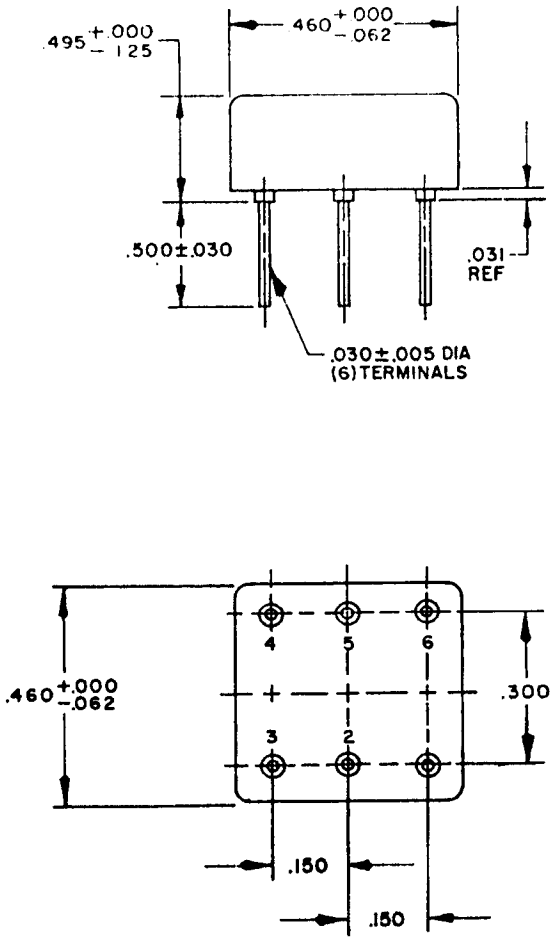
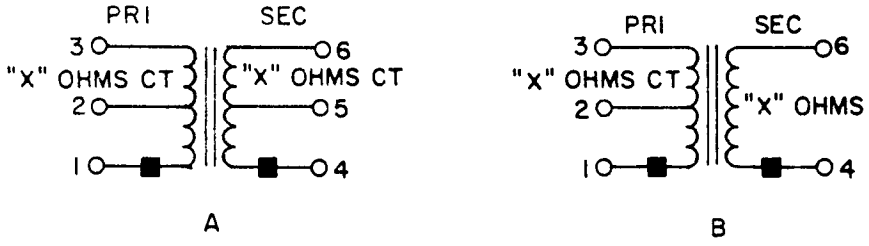


FIGURE 1. Dimensions and configuration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited

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Inches	mm	Inches	mm
.005	0.13	.150	3.81
.030	0.76	.300	7.62
.031	0.79	.460	11.68
.062	1.57	.495	12.57
.125	3.18	.500	12.70



WORKING VOLTAGE: 175 V
ALTITUDE : 70kFT MAX
(SEE TABLE I FOR VALUE OF "X")

CIRCUIT DIAGRAMS AND MARKING

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Marking shall be on sides and/or top of case.
5. Electrical values shall be marked as specified in table I, as applicable.

FIGURE 1. Dimensions and configurations - Continued.

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-3) are used, they indicate the winding and the extreme terminals of the windings.)

Electrical ratings: See table I.

Working voltage: 175 volts.

Frequency range: 300 hertz to 100 kilohertz (except 300 hertz to 20 kilohertz for M27/71-01 and M27/71-02).

TABLE I. Electrical ratings. 1/

Dash number	Circuit diagram (see figure 1)	Primary impedance ohms (1-3)	Secondary impedance ohms (4-6)	Primary DCR ohms ±25%	Primary power level at 300 Hz mW (max)	Secondary DCR ohms ±25%	Primary unbalanced dc current mA (max)
01	A	100 k CT	1 k CT	2,900	10	60	0
02	A	25 k CT	1,000 CT	2,500	10	140	0
03	A	10 k CT	10 k CT	1,000	30	1,300	1.0
04	A	10 k CT	1,500 CT	1,000	30	165	1.0
05	A	10 k CT	600 CT	1,000	30	65	1.0
06	A	4,000 CT	10 k CT	440	30	1,300	1.5
07	A	1,000 CT	600 CT	100	30	65	3.0
08	B	1,000 CT	4	100	30	0.45	3.0
09	A	600 CT	25 k CT	65	30	3,300	4.0
10	A	600 CT	600 CT	65	30	65	4.0
11	B	600 CT	60	65	30	7	4.0
12	B	600 CT	4	65	30	0.45	4.0

1/ Qualification testing to M27/71-01 and all tests per table VII of MIL-T-27 for M27/71-05 and M27/71-09 shall be sufficient to grant qualification approval to M27/71-01 through M27/71-12.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Encapsulated.

Material: Plastic.

Terminals: Pin (printed circuit).

Material: Nickel iron alloy tin plated.

Diameter: 0.030 ±0.005 inch.

Length: 0.500 ±0.030 inch.

Weight: 5.10 grams maximum.

Altitude: 70,000 feet maximum.

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Operating temperature range: -55°C to +130°C.

Terminal strength: MIL-STD-202, method 211, test condition A, 2 pounds.

Dielectric withstanding voltage:

At sea level: 500 volts rms.

At barometric pressure: 300 volts rms.

Electrical characteristics: See table II.

Frequency response:

Reference frequency: 1 kilohertz.

Polarity: Additive, with terminals 3 and 4 connected.

TABLE II. Electrical characteristics.

Dash number	Frequency response at 300 Hz to 100 kHz ±2 dB Ⓒ			
	Zs (1-3) ohms	Z_L (4-6) ohms	E_L volts	Primary dc current (1-3) mA
01 1/	100 k	1,000	17	0
02 1/	25 k	1,000	3	0
03	10 k	10 k	10	0.5
04	10 k	1,500	10	0.5
05	10 k	600	10	0.5
06	4,000	10 k	5	0.25
07	1,000	600	3	1.5
08	1,000	4	3	1.5
09	600	25 k	2.5	2
10	600	600	2.5	2
11	600	60	2.5	2
12	600	4	2.5	2

1/ Frequency response is 300 hertz to 20 kilohertz ±2 dB.

Marking location: See figure 1.

Part number: M27/71-(dash number from table I.)

Ⓒ Substitution data (information only): This military transformer is substitutable for CAGE 70674, part number 324-6A. This in no way implies that the manufacturer's part is substitutable for the military item. This information is not intended to be complete.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - OS, SH
Air Force - 11, 17, 80
DLA - ES

User activities:
Army - AR, ME
Navy - AS, MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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